



# M-Tech Production & Marketing

Technology | Training | Marketing | Projects

## Week 3 - Day 3 Report

Internship Program — Batch 2026

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| Submitted By: | Ghazi Muhammad Abdullah – AI26025 |
| Department:   | AI / ML                           |
| Submitted To: | Malik Mohazin Zahid               |
| Date:         | 15 July 2026                      |

## 1. Introduction

This report outlines the work completed for Experiment 11: Natural Language Processing (NLP), Transformers & BERT under the AI/ML Internship Program 2026. The experiment focused on teaching how Artificial Intelligence systems understand, process, and generate human language. It also introduced Transformer-based models and BERT, which have transformed modern language processing applications.

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## 2. Objectives

- Learn the fundamentals of Natural Language Processing.
  - Study text preprocessing methods.
  - Understand the Transformer architecture.
  - Learn how BERT performs contextual language understanding.
  - Explore practical NLP applications.
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## 3. Task Description

The experiment involved studying different techniques used for processing textual information. It covered text cleaning, tokenization, stop-word removal, stemming, and lemmatization. The experiment also explained the Transformer architecture and how BERT uses bidirectional learning to understand the meaning of words based on their context.

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## 4. Work Done

During this experiment, I learned the complete workflow of Natural Language Processing, beginning with text preprocessing and ending with language understanding models. I practiced various preprocessing techniques that prepare raw text for machine learning. I also studied the Transformer architecture and learned why it performs better than earlier sequential models. Furthermore, I explored BERT and understood how it captures contextual information from both directions of a sentence, making it highly effective for language-related tasks such as sentiment analysis, chatbots, machine translation, and question-answering systems.

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## 5. Challenges Faced

- Understanding self-attention in Transformer models.
  - Learning contextual language representation in BERT.
  - Differentiating between traditional NLP methods and Transformer-based approaches.
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## 6. Key Learnings

- NLP enables machines to understand and generate human language.
- Text preprocessing improves data quality before model training.
- Transformers process language efficiently using self-attention.
- BERT understands the context of words from both directions.
- NLP is widely applied in virtual assistants, chatbots, translation systems, and search engines.

## 7. Conclusion

This experiment provided a comprehensive understanding of Natural Language Processing and modern language models. I learned how Transformer architecture and BERT have improved the performance of NLP applications. The knowledge gained will support the development of intelligent systems capable of understanding and generating human language.

**Intern Signature:**



**Supervisor:**

**Name:** Ghazi Muhammad Abdullah

**Name:**

**Date:** 15 July 2026

**Designation:**